

7-5 Graph Inequalities

Equation Detective Agency | Standard: 6.EE.8 | Color-by-Number Practice

Learning Goal: I can graph the solutions of an inequality on a number line.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: open circle, closed circle, shade

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Describe the graph of $x > 5$. Work: _____ Answer: _____	2. Describe the graph of $x < 8$. Work: _____ Answer: _____
3. Describe the graph of $x \geq 3$. Work: _____ Answer: _____	4. Describe the graph of $x \leq 10$. Work: _____ Answer: _____
5. Describe the graph of $x > -2$. Work: _____ Answer: _____	6. Describe the graph of $x < -4$. Work: _____ Answer: _____
7. Describe the graph of $x \geq -1$. Work: _____ Answer: _____	8. Describe the graph of $x \leq 0$. Work: _____ Answer: _____
9. Describe the graph of $x > 12$. Work: _____ Answer: _____	10. Describe the graph of $x < 15$. Work: _____ Answer: _____
11. Describe the graph of $x \geq 7$. Work: _____ Answer: _____	12. Describe the graph of $x \leq -6$. Work: _____ Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

7-5 Mystery Picture: Detective Badge Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
open at 5, shade right = red	open at 8, shade left = orange	closed at 3, shade right = yellow	closed at 10, shade left = green
open at -2, shade right = blue	open at -4, shade left = purple	closed at -1, shade right = pink	closed at 0, shade left = brown
open at 12, shade right = gray	open at 15, shade left = teal	closed at 7, shade right = black	closed at -6, shade left = light blue

5	2	1	2	2	2	3	3	3	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
3	5	8	9	7	8	9	9	9	12	9	6
3	2	10	2	2	9	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	2	6
3	6	9	1	5	8	11	7	4	10	9	6
5	8	11	3	6	12	1	10	4	2	11	8
8	8	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	5	8	3	11	8
7	10	3	4	4	2	3	3	4	11	2	2
7	12	12	1	11	11	12	12	12	2	2	11
9	9	9	8	8	8	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	open at 5, shade right	red	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
2	open at 8, shade left	orange	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
3	closed at 3, shade right	yellow	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
4	closed at 10, shade left	green	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
5	open at -2, shade right	blue	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
6	open at -4, shade left	purple	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
7	closed at -1, shade right	pink	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
8	closed at 0, shade left	brown	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
9	open at 12, shade right	gray	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
10	open at 15, shade left	teal	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
11	closed at 7, shade right	black	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.
12	closed at -6, shade left	light blue	Open circle for $<$ or $>$; closed circle for $\leq$ or $\geq$.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.